

Matthew Snelgrove

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matthewSnelgrove
portfolio

Education

University of Toronto, Scarborough | Computer Science | CGPA: 3.72

Sept 2021 – Present

Courses: Data Structures, Algorithms, Software Design, Security, Databases, Operating Systems, Web Development

Experience

SS&C Technologies (IFDS) | Software Developer

Sept 2023 – Apr 2024

- Eliminated **3 hours** of managerial work per week by **automating** component dependency tracking with **internal RESTful APIs**, and visualized the output in graphical and tabular views in a modern **React** web app.
- Ensured the maintainability and extendibility of the project with modular design, **jsDoc code documentation**, and comprehensive wiki entries.
- Reduced **defect response time** by **50%** by rapidly identifying and patching **production bugs** across team apps.
- Accelerated modernization projects by **investigating internal tools** and delivering actionable **user guides** to streamline their integration into existing systems.

Ontario Ministry of Transportation | Junior Technical Analyst

Jan 2023 – Apr 2023

- Developed a **RESTful API** to modernize critical driver/vehicle registration systems with **Java Spring Boot**.
- Cut debugging time by **over 50%** through **automated unit and integration testing** with **JUnit** and **Postman**, achieving **100% code coverage** and preemptively identifying errors caused by pending dependency updates.
- Recovered **~3 hours** per week of developer time by **building and deploying** new versions with **Maven** and **Azure Pipelines** instead of time-consuming and error-prone manual builds.

Projects

Superbiddo | RESTful APIs, PostgreSQL, Google Cloud, Docker, Deployment, Automation, Bash, Security Oct 2024 – Dec 2024

- Implemented core functionality for an auction-based marketplace app by developing a **relational database** using **PostgreSQL**, and a **RESTful API** using **Express** with **YAML documentation** following **OpenAPI specification**.
- Optimized a **Google Cloud VM** to host the app with **Docker**, achieving **100% uptime**, cost efficiency, and secure networking through resource fine-tuning, proactive monitoring and alerts, and stringent network configurations.
- Eliminated **5 hours** of manual deployment time weekly by **automating** consistent **deployments** with a **bash script**.
- Mitigated **OWSAP Top 10 web vulnerabilities** by implementing session-based authentication and access control, salted password hashing, data validation and sanitization, CSRF protection, and HTTPS.

Pintos Operating System | C, Systems Programming, Operating Systems, Optimization

May 2024 – Aug 2024

- Extended a rudimentary **operating system** in **C** to support system calls, indexed and extensible files, and ELF program execution while ensuring stability and performance in a **multithreaded** environment.
- Improved fairness and **responsiveness** by implementing priority donation and a multi-level feedback queue scheduler (MLFQS) to dynamically adjust thread priorities based on CPU usage and workload.
- **Optimized** memory management and **reduced page fault frequency by 30%** by designing a virtual memory system with LRU-approximated page replacement using the clock algorithm.

Stock Portfolio App | PostgreSQL, Database Optimization, Relational Databases, SQL

Aug 2024

- Designed and implemented a **PostgreSQL database** for portfolio management, and **optimized** stock correlation and covariance queries by precomputing the results in a **materialized view**, **reducing query time by 98%**.

Technical Skills

Languages: TypeScript, JavaScript, Java, Python, C/C++, C#, SQL (Postgres), Bash, YAML, HTML, CSS

Frameworks & Libraries: Node.js, Express.js, Spring Boot, React, MongoDB, JUnit, Mocha, Chai, NumPy

Developer Tools: Git, Docker, Google Cloud Platform, Azure, Postman, OpenApi,